

TRAINING-FREE · LOCAL REFINEMENT · FROZEN GENERATOR

Small regions. Finer detail.

Refine the geometry and texture that matter,
while preserving the surrounding 3D asset.

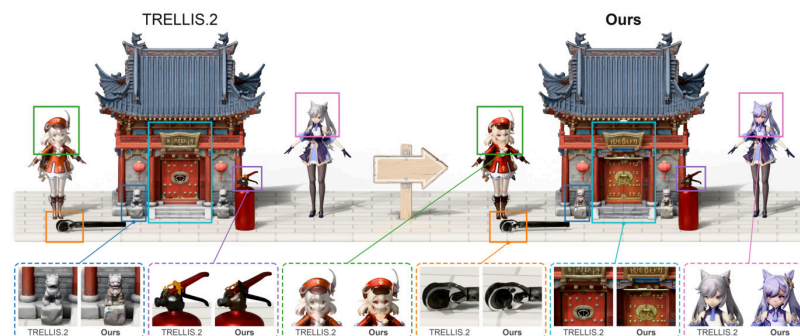
NOFACE3D: Noise-Scheduled Focused Anchoring and Controlled Enhancement for 3D Generation

[Authors and affiliations / anonymous review version]

[Read the paper ↗](#)[Explore results ↓](#)[Code · planned release](#)[BIGDETAIL](#)

A CLOSER LOOK AT GENERATED 3D

PAPER FIGURE 1 · STATIC PREVIEW



Given a generated mesh, its conditioning image, and a selected region, NOFACE3D reallocates the generator's spatial capacity to recover finer local detail.

01 · ADD / DECIDE

Replace this still with a 15-20 second hero loop: full asset → selected region → synchronized detail comparison → full asset. Use a character, a mechanical object, and a thin structure.

Choose final authors/affiliations and review status. Keep code and dataset release labels accurate. The paper button currently opens the supplied draft.

01 / RESULTS **Get closer. See the difference.**

Explore improvements in both appearance and geometry, from character details to compact mechanical structures.

Objects · BIGDETAIL

Characters & general assets

BIGDETAIL comparisons

PAPER PREVIEW

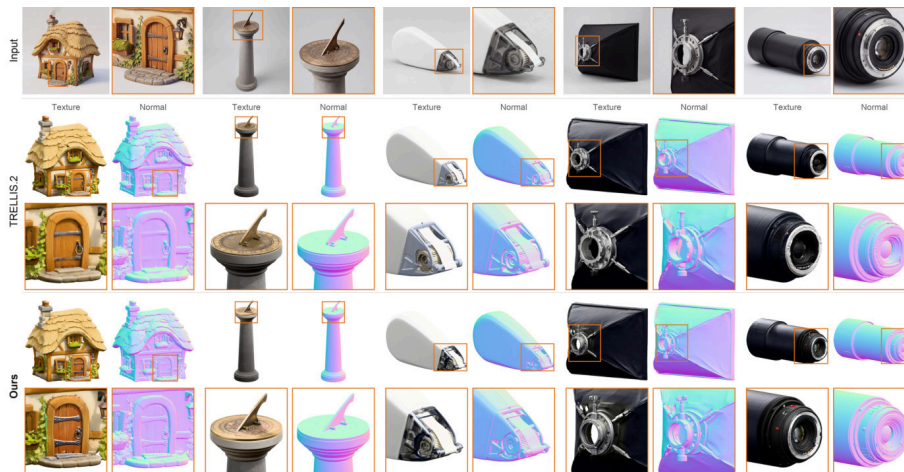


Figure 4 · Input regions, original results, and refinements in textured and normal views.

02 · ADD ASSETS

Start with six strong matched examples. Expand to 8-12 after inspection.

- Character head / hair
- Accessory or thin structure
- Tape-dispenser gear
- Lens-mount contacts
- Two further diverse objects

For each: input image, 2D/3D selection box, original/refined GLBs, camera presets, textured/clay/normal posters, and run metadata.

Check: same lighting, camera, and export settings. Verify side and rear views. Label curated examples.

Interactive comparison — reserved space

Replace the paper preview above with paired 3D assets. Controls below show the intended interface and are not connected.

Original · TRELIS.2

[Original GLB + matching camera]

Refined · NOFACE3D

[Refined GLB + matching camera]

Texture

Clay

Normals

Full asset

Detail

Rotate together

02 / METHOD More capacity, where it matters.

A small region occupies the generator's full input volume. Boundary anchoring keeps regeneration consistent with the surrounding asset.

01

Select a region

Mark a detail-critical region in the mesh and conditioning image.

03

Refine with anchoring

Hold boundary latents close to the base; allow stronger changes farther away.

02

Enlarge & re-encode

Let the selected submesh fill the generator's existing spatial grid.

04

Reintegrate

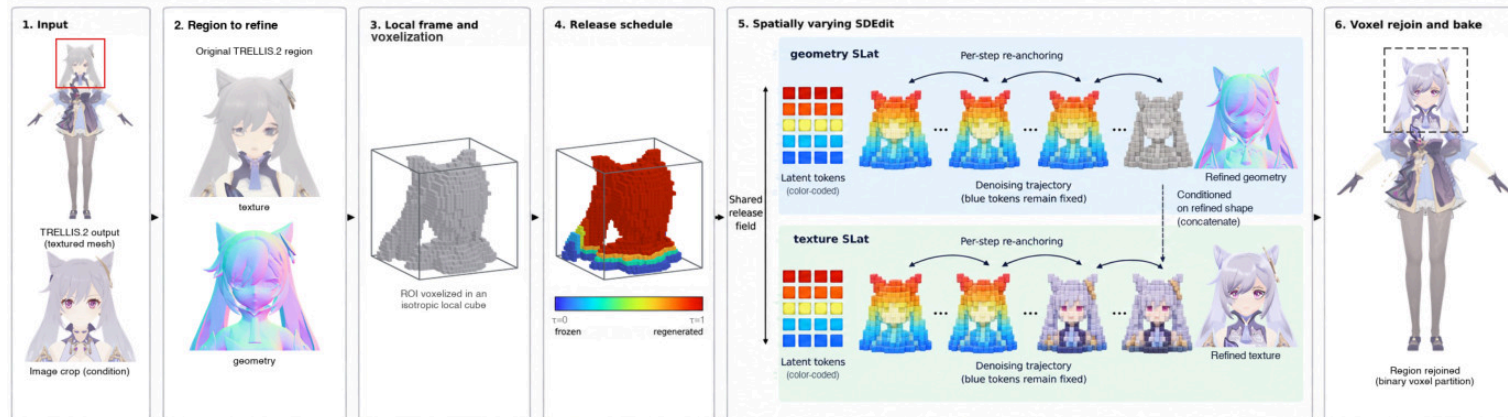
Return the refined region to the original coordinate frame and combine it with the asset.

03 · IMPROVE EXPLANATION

Replace the dense paper diagram with a short four-step animation. Keep the paper figure as an expandable technical view.

Show boundary anchoring visually with an anchored band and graded release field. Avoid equations in the main flow.

Copy check: “training-free” refers to inference on the frozen generator. Do not imply guaranteed exact preservation of the final exported mesh.



03 / ABLATION **The boundary makes the difference.**

Regenerating a crop alone can leave visible discontinuities. Spatial noise scheduling encourages a more coherent transition.



Figure 5 · Without scheduling (top), with scheduling (bottom). Matched detail crops reveal the attachment region.

04 · ADD A DECISIVE COMPARISON

Build a three-column sequence: original / without scheduling / full method.

Use one clearly visible attachment boundary, a fixed camera, and clay plus texture views. Add a second example only if it reveals a different failure.

Use the ablation outputs from the paper. Avoid describing the join as welded or guaranteed seamless.

04 / EDITING A new detail. Then another.

An edited image crop turns the same refinement pipeline into a local 3D editor. Successive edits can target different regions.

Hat → robotic legs

Wings → head refinement

Engine bay → rockets

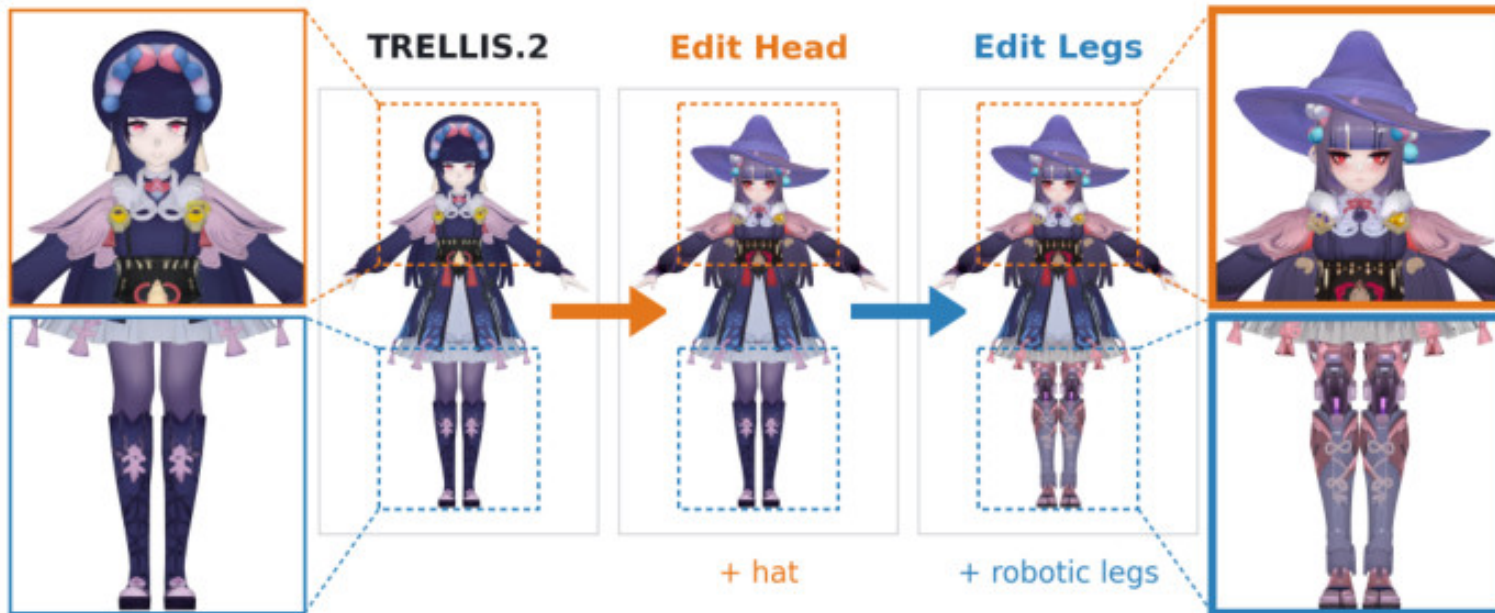


Figure 6 · The second edit targets the legs while retaining the earlier hat edit. Conditioning crops are edited with Gemini Flash Image.

05 · ADD THE EDITING STORY

Show input crop → edited crop → resulting 3D region alongside each step.

Replace the static sequence with original / edit 1 / edit 2 states sharing a camera. Preserve all three meshes.

Label Gemini-edited conditioning. Keep refinement-only benchmarks distinct from image-upscaled or edited demonstrations.

05 / EVALUATION

Visible detail. Measured preference.

83.3%

Human preference

Of judged ratings preferred NOFACE3D. Including “neither,” 73.3% of all responses preferred our method.

4,892

PAniC-3D assets

Large-scale evaluation against the original TRELIS.2 generations.

339

BIGDETAIL assets

Refined assets with annotated detail-critical regions in images and meshes.

► Study context and selected quantitative results

06 · VERIFY BEFORE RELEASE

Values are transcribed from the supplied draft, Tables 2 and 4 and Appendix B.

Reconcile these with the final paper after reruns. Preserve denominators and sample counts; do not turn judge win shares into human preference claims.

Add measured end-to-end runtime only after benchmarking the complete refinement/export pipeline.

06 / BENCHMARK

Meet BIGDETAIL.

339 REFINED ASSETS

Small parts. A broad set of objects.

A benchmark built around large, smooth forms with compact, detail-rich regions: mechanisms, fittings, connectors, and more.

Conditioning images

Generated object images and corresponding detail crops.

Region annotations

Image rectangles and oriented 3D selection boxes.

Evaluation protocol

Paired comparisons of original and refined assets; no artist ground-truth meshes.

[Dataset · planned release](#)[Preview examples ↑](#)

07 · DEFINE THE RELEASE

Add the final dataset landing page, license, archive structure, annotation schema, and evaluation instructions.

Confirm which images, meshes, and outputs are included. The draft states code and BIGDETAIL will be released upon acceptance.

A searchable gallery is optional for a later version. Start with a compact curated preview.

Where refinement still struggles

The method inherits the frozen generator's limitations. Printed labels and lettering may remain distorted after refinement.

ADD AN HONEST EXAMPLE

Insert a matched before/after crop showing unresolved lettering. Label what improved and what did not.

Read, reproduce, build on it.

- Paper ↗
- Code · planned release
- Dataset · planned release

► Citation — metadata pending

08 · FINISH THE PUBLIC PAGE

Add permanent paper, code, and dataset URLs; official citation; acknowledgments; and asset credits.

Use stable hosting for images and meshes. Replace temporary or expiring links. Remove editorial notes and unconnected viewer controls for publication.

First-release checklist

The smallest complete version: one hero, six paired examples, one boundary ablation, and one editing sequence.

- ☐ Select and validate six paired examples
- ☐ Produce the hero loop and mobile poster
- ☐ Add ablation meshes and boundary close-ups
- ☐ Confirm final metrics, credits, and author information
- ☐ Export matching poster images and lightweight GLBs
- ☐ Connect the synchronized comparison viewer
- ☐ Add edited image crops and sequential edit meshes
- ☐ Publish stable resource URLs and verify mobile behavior